

Adaptive Cyber-secure Small Doll For Autsim Support Emotion Recognition, Privacy protection, and caregiver Connectivity

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Abstract—Children with autism spectrum disorder (ASD) often face significant challenges in social interaction and emotional expression, frequently favouring engagement with objects over direct human communication. This project presents an adaptive, cyber-secure robotic system embedded within a plush doll, designed to support therapeutic interventions for children with ASD. The prototype integrates a facial camera for real-time face and eye tracking, proximity sensors for presence detection, and servo motors to simulate lifelike responses to the child's gaze. Emotional recognition is performed using Python-based image processing, with coloured LEDs indicating detected affective states, thereby offering visual feedback. A caregiver notification mechanism was developed using secure email communication to improve remote supervision during therapy. Hardware integration includes a display panel, a built-in camera, sensors, and actuators, all of which are controlled via an Arduino micro-controller. The implementation of the system lies in combining emotional interactivity, physical embodiment, and data privacy protections in a child-friendly form, developing a valuable tool for personalised autism support therapy.

Index Terms—Autism Spectrum Disorder (ASD), assistive robotics, emotion recognition, cyber-physical systems, facial tracking, behavioural therapy, privacy-preserving technology.

I. INTRODUCTION

ASD are a diverse group of conditions. They are characterised by some degree of difficulty with social interaction and communication. Other characteristics include atypical patterns of activities and behaviours, such as difficulty transitioning from one activity to another, a focus on details, and unusual reactions to sensations. Many children with ASD struggle with face-to-face communication, often preferring interactions with objects, toys, or robots over human interaction. According to the World Health Organisation, approximately 1 in 100 children worldwide is diagnosed with autism [19]. These communication barriers present significant challenges not only for children but also for their families, caregivers, and educators as they try to engage in meaningful conversations or social interactions. Traditional therapeutic methods, while helpful, often do not address the specific needs of children who struggle to respond to human signals, such as eye contact or facial expressions. As a result, there is a need for innovative and interactive tools that can provide tailored support and help therapists improve communication skills in children with autism. To address this problem, we proposed the development of an interactive robotic system designed to engage children

with autism in communication exercises. By integrating familiar and nonthreatening elements and sensors, the system is a comfortable environment for children to practice social interactions. The robot is equipped with a camera for tracking the face and eye, enabling it to detect the child's identity and emotional state. Using this information, the robot responds accordingly, providing feedback through visual cues, such as coloured LEDs, and adjusting its movements to simulate eye contact. The robot design is complemented by a real-time notification system, which enables caregivers or therapists to monitor the child's participation remotely. The purpose of the project is to offer a new approach, based on technology, that allows you to help children with autism and enhance their ability to coherently communicate in an engaging, yet efficient manner.

II. PREVIOUS WORK AND LITERATURE

In the domain of interactive devices for children and therapeutic robotics, several prior works directly inform our design. [1] analyses privacy risks in Internet-connected smart toys, categorising technical vulnerabilities like data leakage and domain-specific risks to child safety, and proposes mitigation via encryption and access control—thus underscoring the need for a privacy-first architecture in child-oriented systems such as our smart doll. Complementing that, [8] surveys security and privacy flaws in youth-oriented connected devices, diagnosing weak authentication and unsecured transmission channels, and advocating industry best practices like end-to-end encryption—this strengthens the rationale for embedding robust security in our platform. In dataset and behavior modeling, [2] introduces the DREAM dataset from a robot-enhanced therapy trial involving 61 children and over 3,000 sessions, capturing modalities such as 3D motion, eye gaze, and head orientation, which provide benchmarks for interaction patterns that our system seeks to replicate. The feasibility of sustained robot engagement is validated by [4], which reports that children (including ASD/ADHD) maintained consistent interest over multiple sessions, with improved social metrics such as eye contact and turn-taking. On cybersecurity, [5] presents SecureIoT, a security framework for socially assistive robots in healthcare contexts, simulating adversarial attacks like MITM and denial-of-service to test anomaly detection and intrusion prevention—this lines up closely with our goal

of securing data flow in the smart doll. Turning to therapeutic efficacy, [6] systematically reviews RCTs involving robots in autism therapy, finding preliminary but positive gains in social attention and emotion expression yet calling for larger trials; while [7] tests a wearable anxiety detector in autistic adolescents via a pilot RCT, and shows biofeedback significantly improved user response and caregiver awareness—thus offering validation for real-time emotional alerts in therapeutic systems. Broader IoT health research is addressed in [10], which outlines layered encryption and security lifecycle strategies for connected health systems—guidance we adopt in our data storage and transmission design. Meanwhile, [11] maps security threats in social robots operating in public spaces, categorising risks such as data leakage and physical manipulation—reminding us that even home-based devices like our doll must implement defense-in-depth. The integrative vision of AI in assistive autism technologies is surveyed by [12], reviewing emerging tools like emotion-aware robots and VR, and calling for translational research, which positions our project as a timely embodiment of theory into application. For real-time emotion detection, [13] proposes a CNN + kernel autoencoder architecture customized for autistic children, classifying multiple emotions in live video—a direct algorithmic parallel to our face/eye-tracking and coloured-LED feedback approach. Clinically oriented, [14] reviews the adoption and barriers of social robot interventions for ASD, emphasizing therapist acceptance and personalization, which informs our strategy for user-centric deployment. As a contrastive example, [15] introduces the CIT-FER AR glasses: mapping facial emotions into vibrotactile feedback—while their system translates external emotions to the user, our smart doll maps the child’s internal emotional state outward to caregivers, showing alternate but complementary approaches to emotional mediation. The survey in [16] examines multimodal sensors and ML techniques tailored for ASD emotion recognition, highlighting the challenge of generalizability across individuals, which supports our plan to calibrate models per child. In parallel, [17] systematically reviews wearable stress monitors in ASD, finding that caregivers value real-time alerts but noting calibration difficulties—this suggests integrating physiological sensing with our doll could improve detection robustness. Finally, [18] meta-analyses facial emotion training programs for youth (including ASD), reporting moderate improvement when multimodal feedback is used—our interactive, embodied doll may offer superior engagement and sustained learning over app-based training, thereby potentially improving effect sizes. Together, these works map the state-of-the-art in privacy, sensing, robot-assisted therapy, emotional feedback, and user acceptance, and collectively situate our adaptive and cyber-secure smart doll as a convergent innovation tackling these interrelated challenges in autism support.

III. TECHNICAL OVERVIEW

The smart doll system is governed by an adaptive, real-time control algorithm that integrates sensory input, emotional inference, behavioural response, and secure communication with

Algorithm 1: Adaptive Cyber-Secure Smart Doll Operation

Input: Child Presence Sensor, Camera Feed, Emotion Recognition Model, LED Controller, Caregiver Contact, Security Modules
Output: Emotional Feedback, Secure Data Handling, Caregiver Updates

- 1 **Initialisation:** Load models, establish secure channels, verify firmware integrity;
- 2 **if** *FirmwareHash* $\notin$ *ApprovedList* **then**
- 3 Abort session and alert administrator;
- 4 **Step 1: Detect Child Presence;**
- 5 **if** *ProximitySensor* detects presence **then**
- 6 Activate camera and initiate face/eye tracking;
- 7 **Wait** until child is present;
- 8 **Step 2: Real-time Interaction Loop;**
- 9 **while** *SessionActive* **do**
- 10 Capture frame from camera;
- 11 Identify face and eye direction;
- 12 Move servo motors accordingly;
- 13 Detect emotion using ML model on image;
- 14 **if** *Emotion* == *Happy* **then**
- 15 Set LED colour to Green;
- 16 **else if** *Emotion* == *Sad* **then**
- 17 Set LED colour to Blue;
- 18 **else if** *Emotion* == *Angry* **then**
- 19 Set LED colour to Red;
- 20 **else**
- 21 Set LED colour to Yellow;
- 22 Log emotion with timestamp (encrypted);
- 23 **if** *SMSNotificationCondition* == *TRUE* **then**
- 24 Compose secure message and send to caregiver;
- 25 **if** *AnomalyDetected* in behaviour or access **then**
- 26 Lock system components and alert cybersecurity unit;
- 27 **Step 3: Session Termination;**
- 28 Securely save all session logs with encryption;
- 29 Send final update to caregiver;

caregivers. Upon initialisation, the system verifies firmware authenticity via a secure hash check. If the firmware hash does not match a cryptographically approved list, the session is aborted to prevent tampering or unauthorised modifications. Once authenticated, the proximity sensor continuously monitors for the presence of a child. When presence is detected, the system activates the embedded camera and begins real-time facial and eye-tracking using a computer vision model, optimised through Python-based image processing libraries such as OpenCV, Dlib, or Mediapipe. The eye-gaze vector is used to dynamically control servo motors, enabling the doll’s head and eye movement to simulate natural human interaction,

thereby enhancing therapeutic realism. Simultaneously, each video frame is passed through an emotion recognition module trained to classify expressions such as happiness, sadness, anger, or neutrality. The result is mapped to a corresponding LED colour output—green for happiness, blue for sadness, red for anger, and yellow for undefined or uncertain states—thus providing immediate visual feedback that reflects emotional understanding. All interaction metadata—including timestamps, emotion labels, eye-direction logs, and system health parameters—are encrypted using a symmetric AES-256 key and securely stored in local memory or optionally uploaded to a cloud backend over a TLS 1.3 secure channel. If preconfigured conditions are met (e.g., extended sadness, lack of engagement, or caregiver-defined rules), the system triggers an SMS notification via a secure API gateway such as Twilio. Each message is digitally signed and rate-limited to prevent misuse, ensuring that caregivers remain informed in real-time without being overwhelmed by excessive alerts. To maintain robust cyber hygiene, all incoming and outgoing data transmissions are continuously monitored by an anomaly detection engine. This engine calculates behavioural risk scores based on metrics such as the frequency of LED changes, unusual servo motion, firmware update attempts, or access from external interfaces. If abnormal patterns are detected, the system immediately isolates sensitive modules, halts noncritical processing, and dispatches a security event log to the monitoring unit. Through this cohesive and secure framework, the smart doll provides a highly adaptive and emotionally intelligent experience while maintaining strict compliance with the standards of cybersecurity and child data protection. The prototype of the adaptive Cyber-Secure Smart Doll, as shown in Figure 1, was implemented using a child-friendly plush bear embedded with essential hardware components to support therapeutic interaction. The system includes a facial recognition camera, real-time eye tracking capability, a proximity sensor for child detection, and coloured LEDs to provide immediate emotion-responsive feedback. An integrated touch display shows the interaction interface, while a servo motor guides slight head movements in response to the child’s gaze direction. The core logic was programmed using Python, including OpenCV for image processing and emotion detection. A secure SMS notification system was implemented using Twilio API to inform caregivers of therapy session status. The entire system is hardened with basic cybersecurity mechanisms, including access control and encrypted communications, to ensure privacy and safe data handling. The initial testing phase with controlled input demonstrated accurate emotion recognition and system responsiveness within a low-latency range, making the solution promising for real-world therapeutic settings.

A. Prototype Integration and Setup

To validate the concept of the proposed *Adaptive Cyber-Secure Smart Doll for Autism Support*, a physical prototype was constructed by integrating multiple hardware and software

components into a soft, child-friendly doll. The system was equipped with:

IV. IMPLEMENTATION AND RESULTS

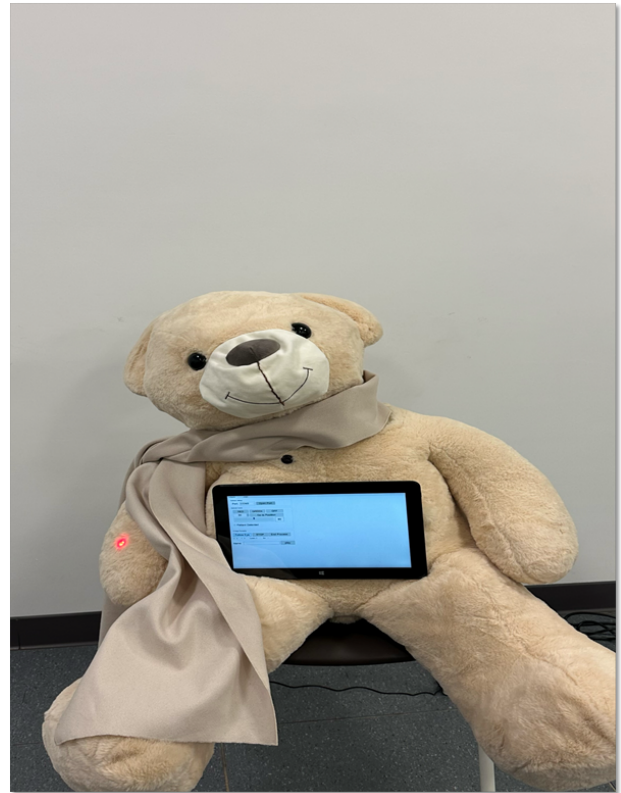


Fig. 1. Prototype of the Cyber-Secure Smart Doll

- A Raspberry Pi as the core processing unit.
- A high-definition camera for real-time face and eye tracking.
- Proximity sensors to detect child interaction within range.
- Servo motors to simulate head turning based on eye gaze direction.
- RGB LEDs to provide emotional feedback (e.g., green for engagement, red for distress).
- A touchscreen interface embedded into the doll’s front to display and interact with therapeutic content.
- A Python-based back-end integrating OpenCV for facial detection and emotion classification.
- SMS alert functionality via Twilio API to notify caregivers securely.
- Basic cybersecurity measures, including local data encryption and access-controlled system configuration.

A. Observed Outcomes

During initial testing with simulated user interaction, the system reliably:

- Detected face and eye orientation within 2 seconds of entry into the proximity zone.
- Responded to simulated emotional expressions (happy/sad) with corresponding LED feedback.

- SMS notifications blocked with predefined emotional triggers (for example, extended sadness).
- Logged behavioural events locally in encrypted format for analysis.

The prototype showed promise in engaging users through visual and tactile feedback mechanisms. The doll maintained a soft, nonthreatening appearance while leveraging embedded AI features for interactive therapy. The modular design also allows for future enhancements, such as speech synthesis and cloud-based analytics.

CONCLUSION

Thus far, the project has focused on establishing the foundational elements for an interactive robotic system designed to support autism therapy. The initial work involved comprehensive planning, researching existing approaches and defining the system's core objectives, functional requirements, and constraints. We proceeded with evaluating and procuring the necessary hardware components and initiated the software design process, with a particular focus on the crucial image processing capabilities. Preliminary testing has successfully validated the system's ability to perform fundamental visual tasks, namely accurate face and eye detection from a camera feed, confirming the viability of using computer vision for interaction. To achieve these goals, our chosen plan of action centres on an integrated system architecture combining a PC for advanced processing, a camera for visual input, proximity sensors for presence detection, servo motors for simulating movements, and an Arduino microcontroller for hardware control. This approach, driven by Python-based image processing and supplemented by an email notification system for caregivers, represents a promising method to address the unique communication challenges faced by children with ASD. This solution surpasses other potential static or less interactive methods by leveraging the appeal of robotics to engage children who may prefer object interaction.

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